

Using AWS Certificate Manager

(LAB-M06-02)

Overview

In this lab, you will be accessing your application using secure communication. You are installing SSL/ TLS certificate in Load balancer to access it securely.

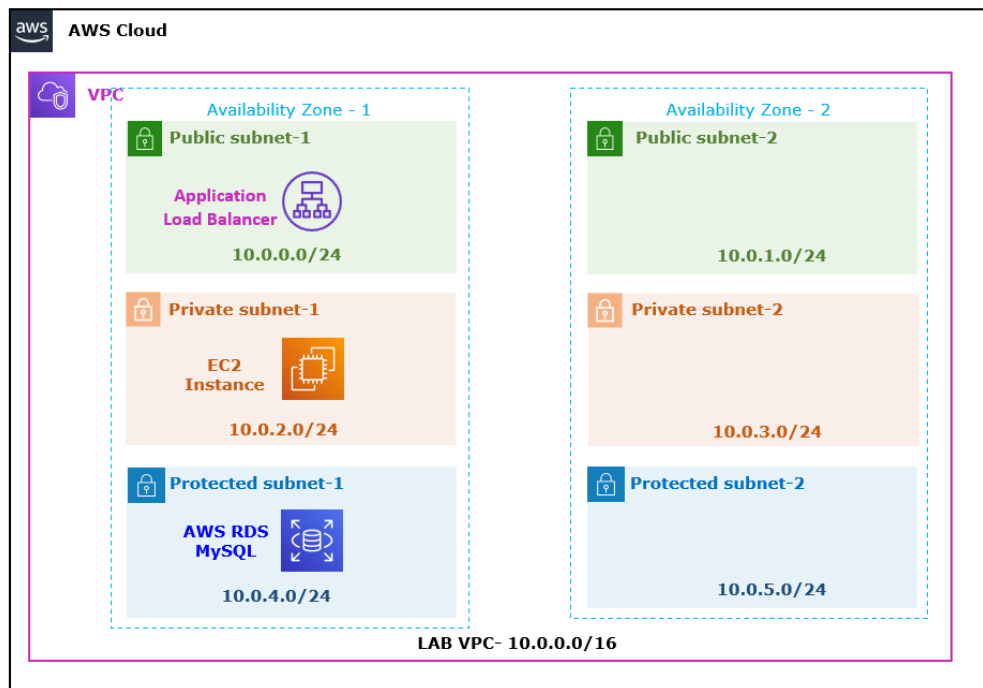
Objectives

After completing this lab, you will be able to:

- Registering Domain name.
- Create the Route 53 (DNS Service)
- Creating SSL/ TLS certificate using AWS Certificate Manager (ACM).
- Uploading SSL/ TLS certificate to Load balancer.

Task 1: Configure the Domain Name

In this task you are registering the Domain name.



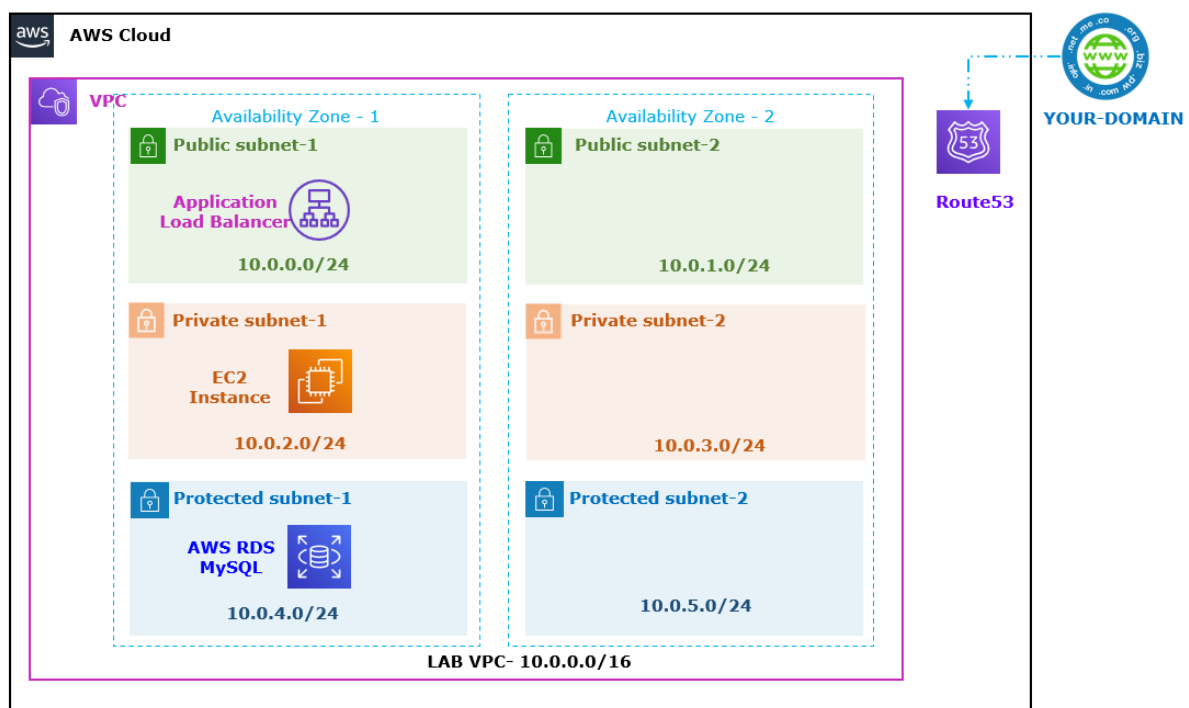
Step 1: Register the Domain Name

1. **Register** the **Domain Name**.

Note: Refer the **Register Domain Name.txt manual** to register domain name, provided with the Lab manual.

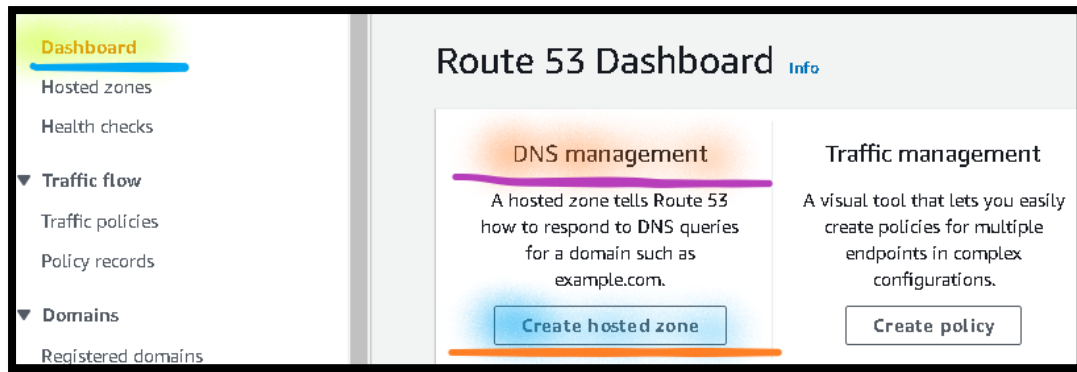
Task 2: Create DNS Services

In this task you are creating the DNS services with the registered domain name.



Step 1: Create a public hosted zone

2. In the **AWS Management Console**, on the **Services** menu, click **Route53**.
3. Click **Dashboard**.
4. Select **Create hosted zone** under **DNS management**.



a. In the **Hosted zone configuration** section:

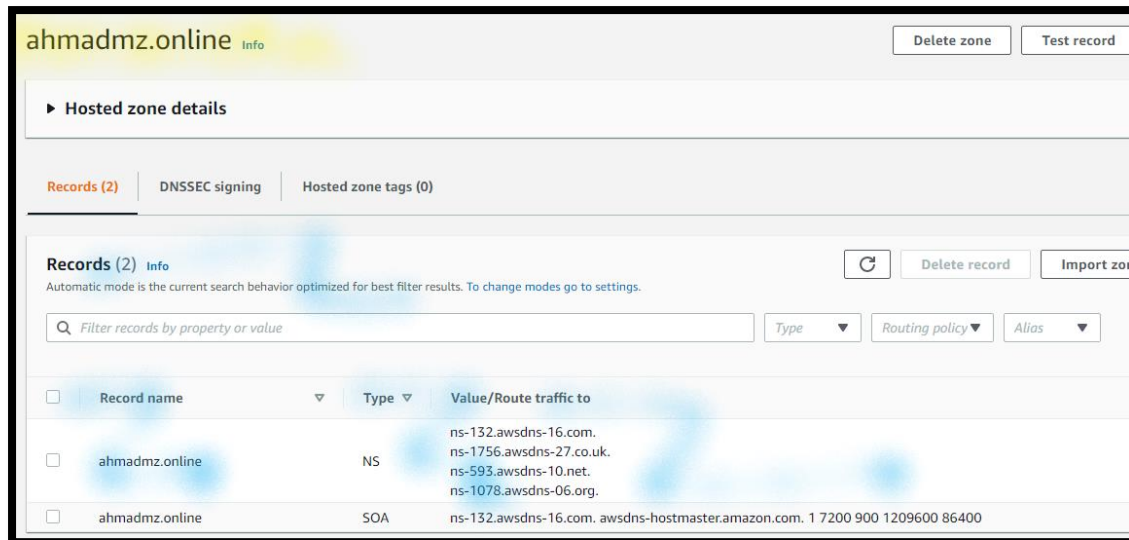
- i. **Domain name:** Write the **name of the domain** (that you register it in the previous step).

Note: Provide domain name only (like **ahmad.com**).

- ii. **Description – optional:** Write **Public DNS Server**.
- iii. **Type:** Select **Public Hosted Zone**.

- iv. Select **Create hosted zone**.

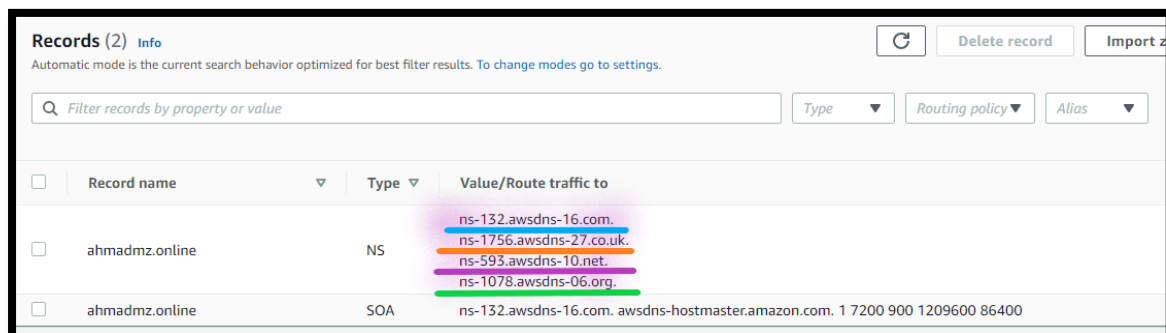
Note: You can see the domain details.



Step 2: Copy the Name servers for a Hosted zone

5. From the **Hosted Zone** console:

- Select **Records**.
- Copy **all the four Name servers** in the **Notepad**.

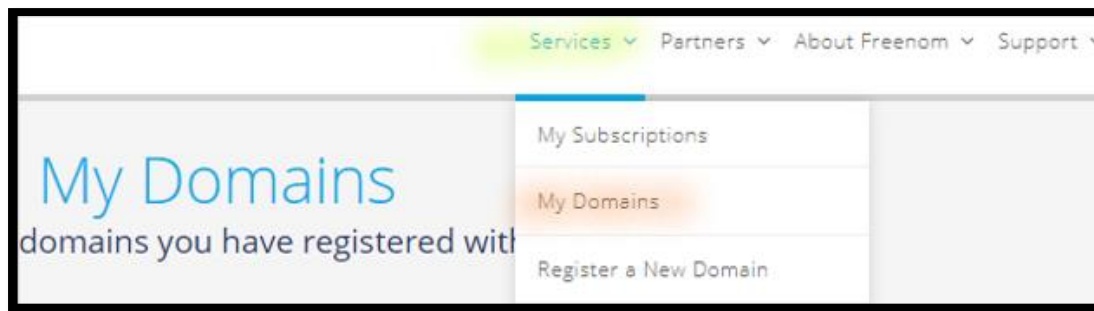


Step 3: Configure Name servers for a Domain name

Update the Name Service in Freenom [Domain Registrar]

Note: If you registered the Domain name in **Hostinger**, Follow the **Line number 7**.

- Login** into **Freenom.com**.
 - Select **Services**.
 - Select **My Domains**.



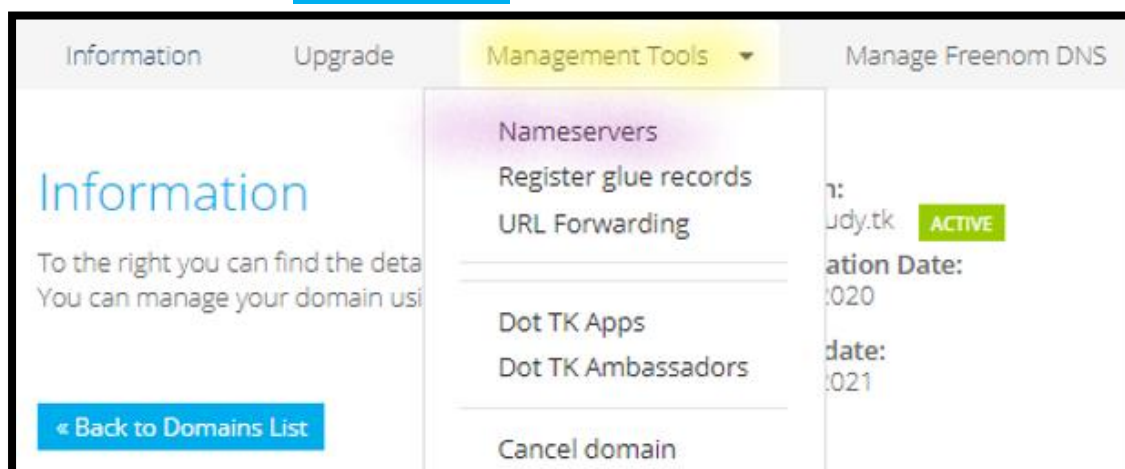
Note: You can see your **Domain name**.

- c. Select **Manage Domain**.

Domain	Registration Date	Expiry date	Status	Type	
cloudstudy.tk	2020-04-29	2021-04-29	ACTIVE	Free	Manage Domain

- d. Select **Management Tools**.

- e. Select **Nameservers**.



- f. Select **Use custom nameservers**.

- Update** the **all the four Name servers** (which you have copied in the previous step).
- Select **Change Nameservers**.

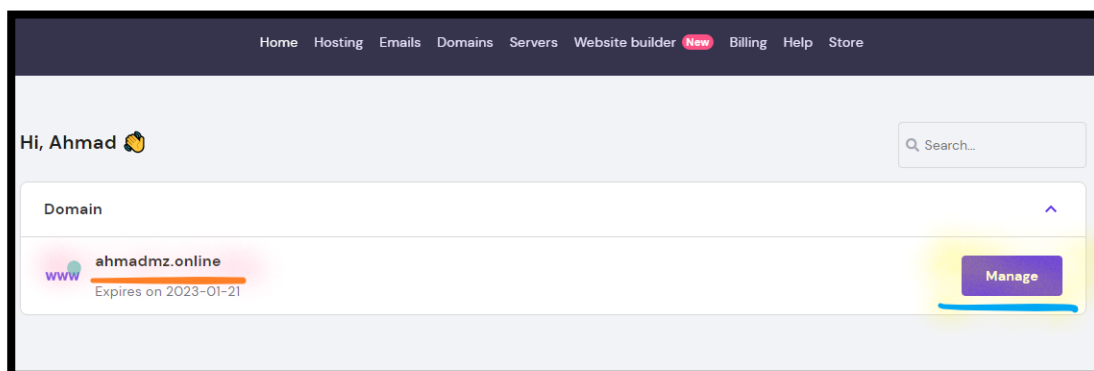


Note: Usually, it take **up to few hours** before name server changes are distributed. But you can try to test the same.

Update the Name Service in Hostinger [Domain Registrar]

Note: If you registered the Domain name in **Freenom**, **Follow** the **Line number 6**. After **Line number 6**, got to **Task 2**.

7. **Login** to **Hostinger**.
 - a. Select **Manage** (against you domain name).



- b. Select **Change** against **Nameservers**.

ahmadmz.online - My domains - ahmadmz.online

Domain Information

Status ✓ Active

Email verification status ✓ Verified

Expires at 2023-01-21 05:29:59

Secret key i ***** 👁

Nameservers

ns1.dns-parking.com

ns2.dns-parking.com **Change**

- i. **Update** the **all the four Name servers** (which you have copied in the previous step).

ahmadmz.online - My domains - ahmadmz.online - DNS / Nameservers

Nameservers

Nameservers handle internet requests for your domain. You can use Hostinger nameservers or use custom nameservers to point to other hosting provider.

ns1.dns-parking.com

ns2.dns-parking.com

Select Nameservers

☐ Use Hostinger nameservers (recommended)

☒ Change nameservers

ns-1221.awsdns-24.org

ns-574.awsdns-07.net

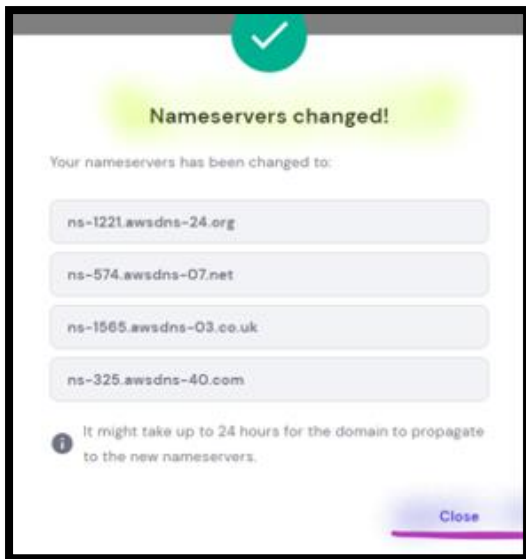
ns-1565.awsdns-03.co.uk

ns-325.awsdns-40.com

Save **Cancel**

- ii. Select **Save**.

Note: You can see the message **Nameservers changed**.

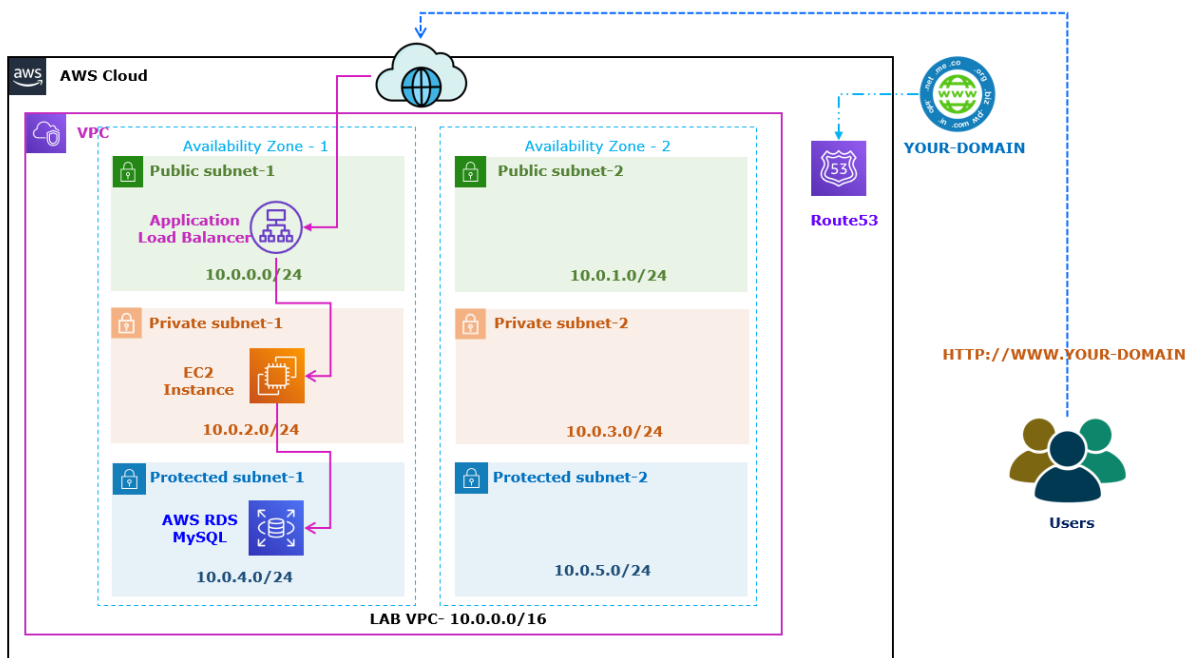


Note: Usually, it take **up to few hours** before name server changes are distributed. But you can try to test the same.

c. Select **Close**.

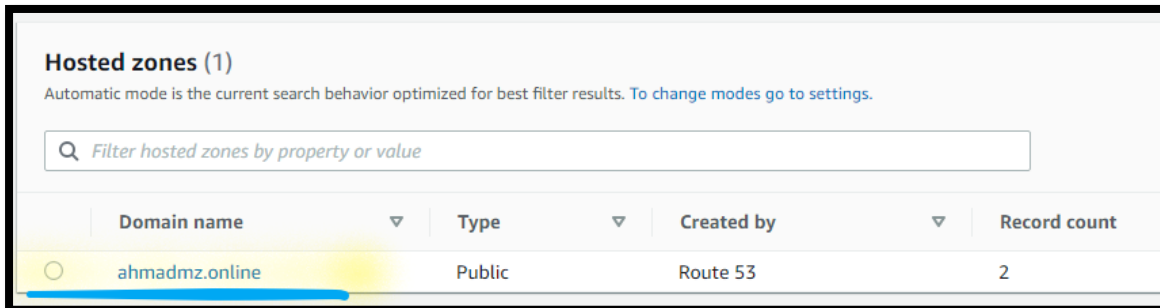
Task 3: Access the Load Balancer with the Domain Name

In this task you are creating the DNS record set to access your application using domain name.

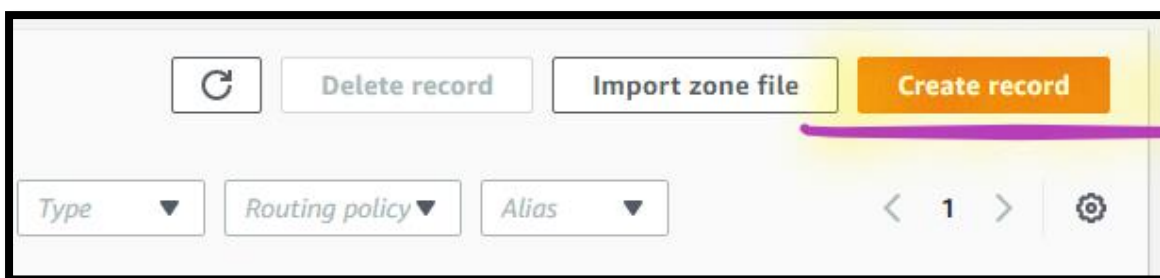


Step 1: Create Record for Load Balancer

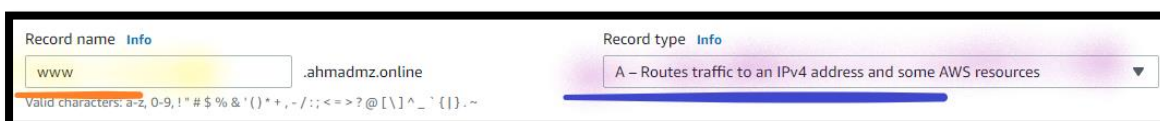
8. In the **AWS Management Console**, on the **Services** menu, click **Route53**.
9. **Go to the left**, select **Hosted zones**.
 - a. Open your **Domain name**.



- i. Select **Create record**.

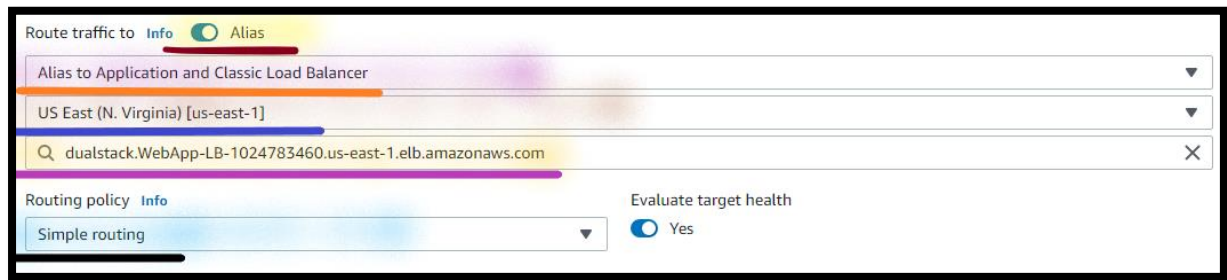


- a) **Record name**: Write **www**.
- b) **Record type**: Select **A – Routes Traffic to IPv4 address and some AWS resources**.



- c) **Alias**: **Enable** the **Alias**.
- d) **Route traffic to**:
 - 1) **Choose an endpoint**: Dropdown and select **Alias to Application and Classic Load Balancer**.
 - 2) **Choose Region**: Dropdown and Select **US East**.
 - 3) **Choose load balancer**: Dropdown & select the **WebApp-LB** load balancer.

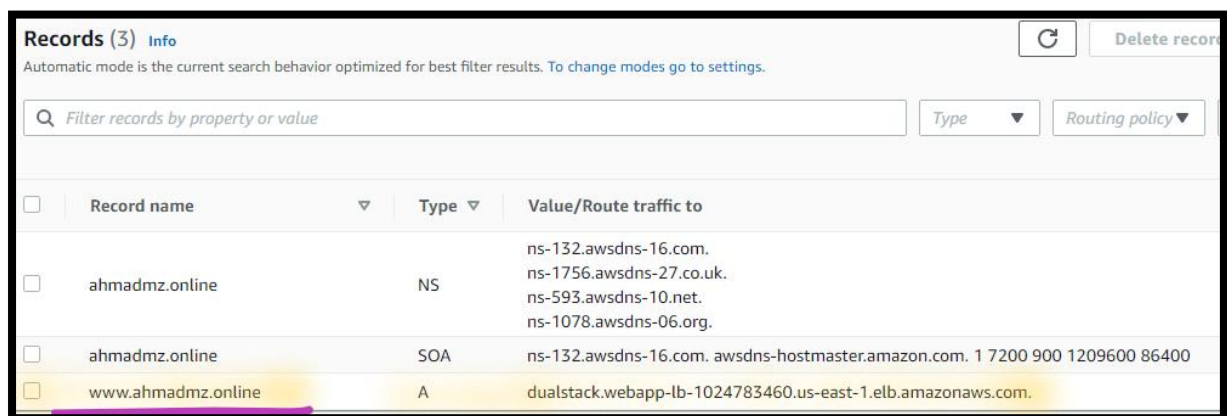
- 4) **Routing policy:** Dropdown and select **Simple routing**.



- e) Select **Create records**.

Step 2: Copy the FQDN

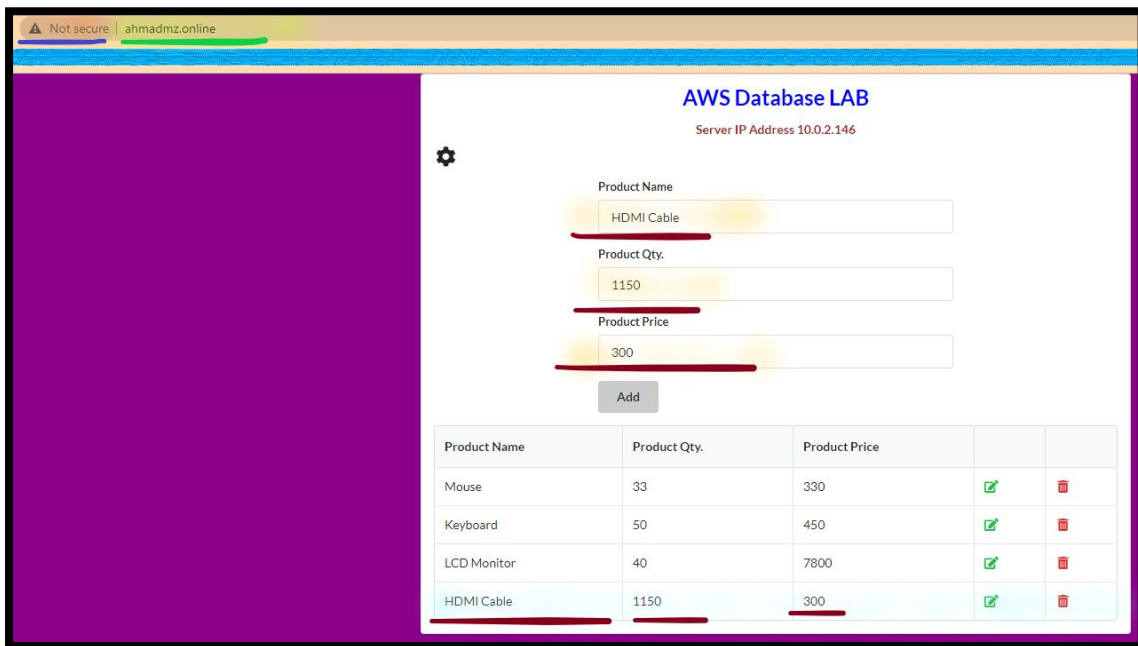
10. From, **Domain name Hosted zones**.
 - a. **Copy** the **www** record name ((like www.ahmad.com)).



Step 3: Access the Load Balancer using FQDN

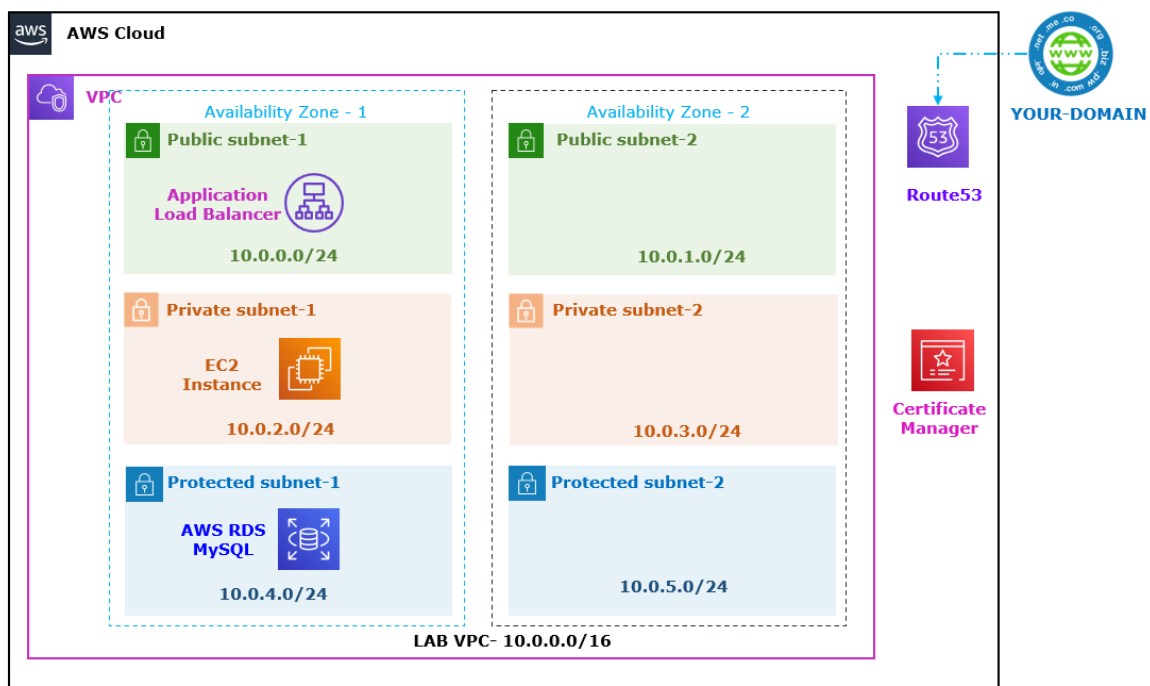
11. From your **local Desktop/Laptop Web browser**, type **FQDN** of **www record**.
 - a. Open the **Settings** (**gear icon** in the web page).
 - b. You get the **Prompt** to enter the **Connection String**.
 - i. **Type Connection string** (which you have created in the previous step).
 - ii. Select **Ok**.
 - c. **Add additional Product inventory** details.

Note: You can see the site is showing as **Not secure**.



Task 4: Generate certificate

In this task you are generate the SSL/ TLS certificate for your application domain name.



Step 1: Generate Certificate

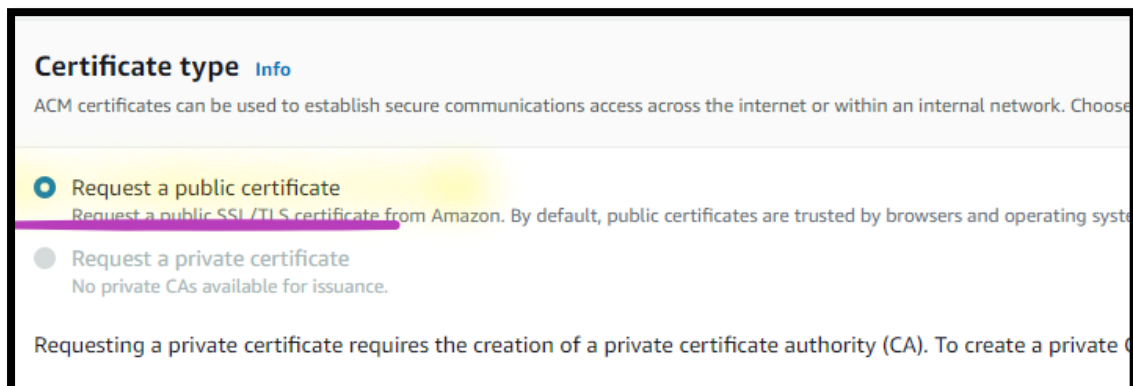
12. In the **AWS Management Console**, on the **Services** menu, click **Certificate Manager**.

13. Select **Request a certificate**.



a. In the **Certificate type** section:

i. Select **Request a public certificate**.



ii. Select **Next**.

b. In the **Domain names** section:

i. **Fully qualified domain name:** Write **FQDN** of **www record**.

ii. **Select validation method:** Select **DNS validation - recommended**.

Domain names

Fully qualified domain name [Info](#)

Add another name to this certificate

You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" to reach your site by either name.

Select validation method [Info](#)

Select a method for validating domain ownership

☒ DNS validation - recommended
Choose this option if you are authorized to modify the DNS configuration for the domains in your certificate request.

☐ Email validation
Choose this option if you do not have permission or cannot obtain permission to modify the DNS configuration for the domains in your certificate request.

c. Select **Request**.

Note: You can see the **Certificate** with **Pending validation**.

Note: If you can not see the Certificate, **Refresh** your screen.

Certificates (1)					
☐	Certificate ID	Domain name	Type	Status	In use?
☐	c0a79dfb-9745-40ab-828b-1afe92d789a6	www.ahmadmz.online	Amazon Issued	Pending validation	No

Step 2: Validate the Certificate

14.From the **Certificate console**:

- a. Open the **Certificate ID**.
 - i. Select **Create records in Route 53**.

Create records in Route 53		Export to CSV
		< 1 >
CNAME name	CNAME value	
 _5637602a19018623aab434a09deb1039.www.ahmadmz.online.	 _e050a2a1b0eb0b57a54161dcae80710b.mntkzmhvzg.acm-validations.aws.	

ii. Select **Create records**.

Create DNS records in Amazon Route 53 (1/1)

Q Search domains

1 match

< 1 >

Validation status: Pending validation X

Validation status: Failed X

Is domain in Route 53?: Yes X

Clear filter

☒	Domain	Validation status	Type	CNAME name	CNAME value	Is domain in Route 53?
☒	www.ahmadmz.online	Pending validation	CNAME	_5637602a19018623aab434a09deb1039.www.ahmadmz.online.	_e050a2a1b0eb0b57a54161dcae80710b.mntkzmhvzg.acm-validations.aws.	Yes

Cancel

Create records

Note: You can see the message **Successfully created DNS records**.

Successfully created DNS records Successfully created DNS records in Amazon Route 53 for certificate with ID c0a79dfb-9745-40ab-828b-1afe92d789a6.
AWS Certificate Manager > Certificates > c0a79dfb-9745-40ab-828b-1afe92d789a6 c0a79dfb-9745-40ab-828b-1afe92d789a6 Certificate status

Step 3: Verify the DNS Records

15. In the **AWS Management Console**, on the **Services** menu, click **Route53**.

16. **Go to the left**, select **Hosted zones**.

a. Open your **Domain name**.

Note: You can see the **new record** created for **ACM validation**.

Records (4) Info Refresh Delete record

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Q Filter records by property or value Type ▼ Routing policy ▼ Alias

☐	Record name ▼	Type ▼	Value/Route traffic to
☐	ahmadmz.online	NS	ns-132.awsdns-16.com. ns-1756.awsdns-27.co.uk. ns-593.awsdns-10.net. ns-1078.awsdns-06.org.
☐	ahmadmz.online	SOA	ns-132.awsdns-16.com. awsdns-hostmaster.amazon.com. 1 7200 900 1209600 86400
☐	www.ahmadmz.online	A	dualstack.webapp-lb-1024783460.us-east-1.elb.amazonaws.com.
☐	_5637602a19018623aab434a09...	CNAME	_e050a2a1b0eb0b57a54161dcae80710b.mntkzmhvvg.acm-validations.aws.

Step 4: Check the Certificate Status

17. In the **AWS Management Console**, on the **Services** menu, click **Certificate Manager**.

18. Select **List certificate**.

Note: You can see the **Certificate** with **Pending validation**.

AWS Certificate Manager X

AWS Certificate Manager > Certificates

Certificates (1)

☐	Certificate ID	Domain name ▼	Type ▼	Status ▼	In use?
☐	c0a79dfb-9745-40ab-828b-1afe92d789a6	www.ahmadmz.online	Amazon Issued	Pending validation	No

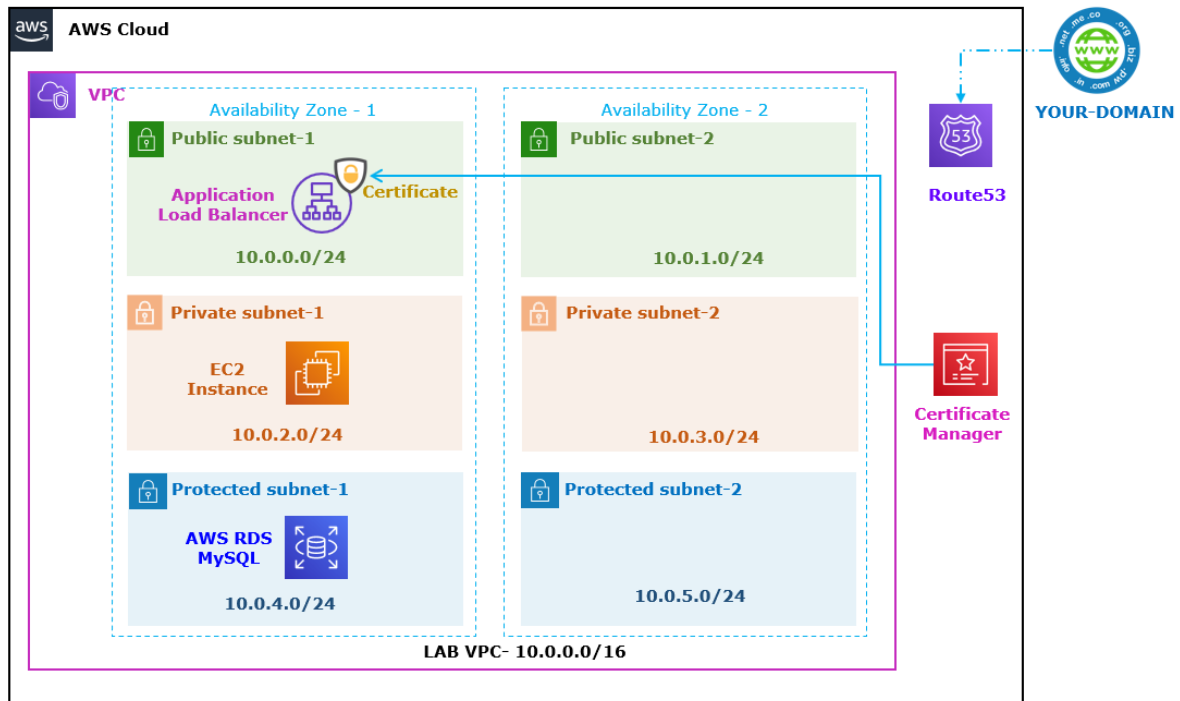
Note: Wait for ~15 mnts, before you can see the **Certificate** with **Issued Status**.

Certificates (1) Refresh Delete

☐	Certificate ID	Domain name ▼	Type ▼	Status ▼	In use?
☐	c0a79dfb-9745-40ab-828b-1afe92d789a6	www.ahmadmz.online	Amazon Issued	Issued	No

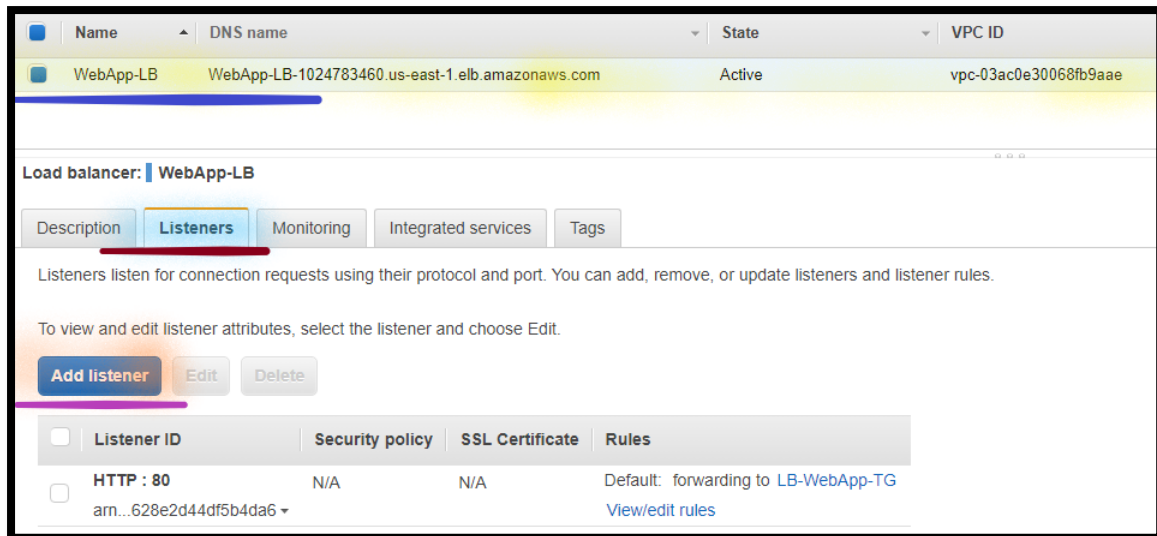
Task 3: Configure Load Balancer

In this task you are uploading the SSL/ TLS certificate to your load balancer.



Step 1: Configure the Listener

19. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
20. **Go to the left**, select **Load Balancers**.
21. Select **WebApp-LB**.
 - a. **Go below**, select the **Listeners**.
 - i. Select **Add listener**.

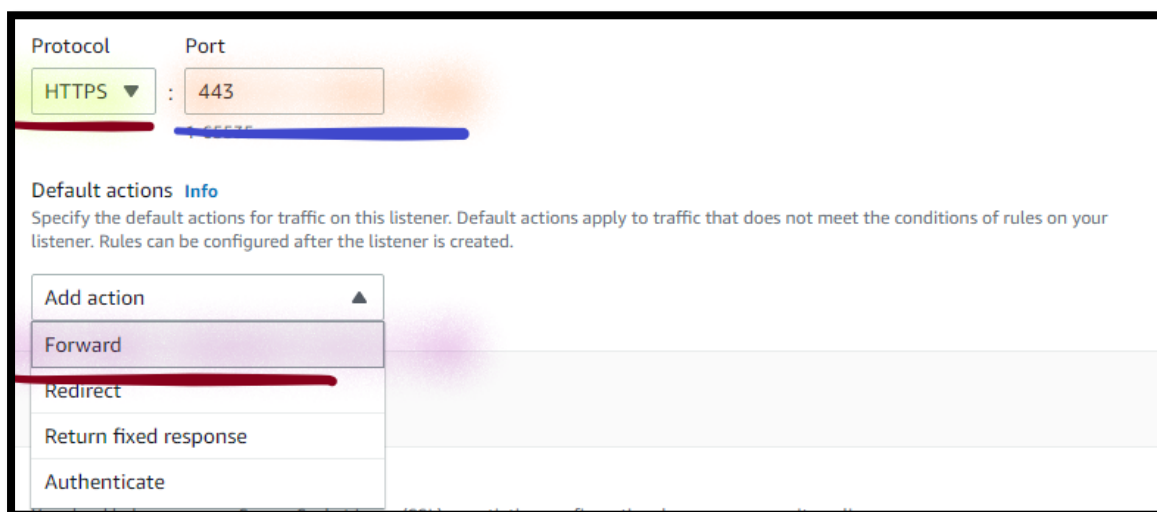


a) In the **Listener details** section:

1) **Protocol**: Dropdown and select **HTTPS**.

I. **Port**: Write **443**.

2) **Default actions**: Dropdown and select **Forward**.



I. **Target group**: Dropdown and select **LB-WebApp-TG**.

▼ 1. Forward to [Info](#) Remove

Target group Weight (0-999)

LB-WebApp-TG HTTP 1 X

Target type: Instance, IPv4

name distribution: 100%

Select a target group 0 X

[Create target group](#)

☐ **Enable group-level stickiness** [Info](#)

If you enable stickiness for your target group, requests routed to it remain in the same group for the duration you specify.

b) In the **Secure listener settings** section:

1) **Default SSL/TLS certificate:**

- I. **First block:** Dropdown and select **From ACM**.
- II. **Second Block:** Dropdown and select **www Certificate** (generated via ACM).

Security policy

Your load balancer uses a Secure Socket Layer (SSL) negotiation configuration, known as a security policy, to negotiate SSL connections with clients.

ELBSecurityPolicy-2016-08

[Compare security policies](#)

Default SSL/TLS certificate

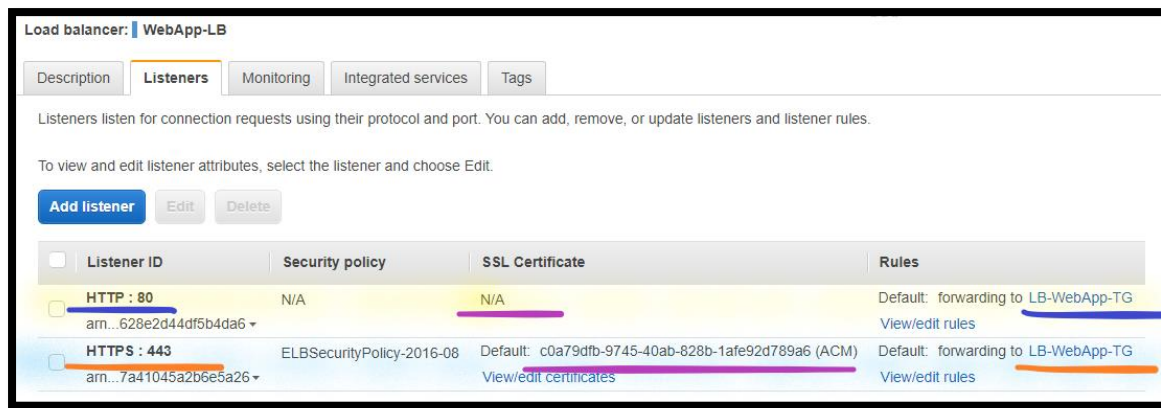
The certificate used if a client connects without SNI protocol, or if there are no matching certificates. This certificate will automatically be added to your listener certificate list.

From ACM www.ahmadmz.online c0a79dfb-9745-40ab-828b-1afe92d789a6 Request new ACM certificate

c) Select **Add**.

d) Select **View Listener**.

Note: You can see the **Two Listeners** [HTTP and HTTPS].



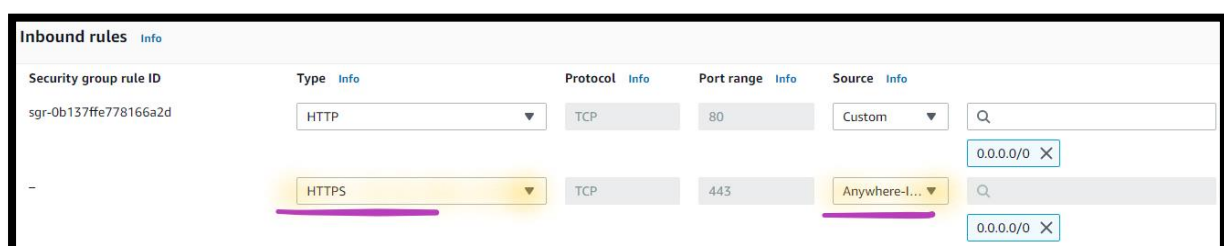
Step 2: Update the Load Balancer Security Group

22. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

23. Select the **Security Groups**.

24. Open the **LB-SG**.

- a. Go below, select the **Inbound**.
- b. Select the **Edit Inbound rules**.
- c. Select **Add rule**:
 - i. **Type**: Dropdown and select **HTTPS**.
 - ii. **Source**: Dropdown and select **Anywhere-IPv4**.
 - iii. **Description**: Write **Allow HTTPS Traffic from Anywhere**.



- d. Select **Save rules**.

Step 3: Update the Load Balancer NACL

25. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

26. Select the **Security Groups**.

27. **Go to the left**, select **Network ACLs**.

a. Open **LB-NACL**.

b. **Go below**, select **Inbound rules**.

i. Select **Edit Inbound rules**.

a) Click **Add new rule**.

1) **Rule number**: Write **120**.

2) **Type**: Dropdown and select **HTTPS**.

3) **Source**: Write **0.0.0.0/0** (anywhere).

Info: You have opened the **Inbound Port 443**, which allows traffic from anywhere to Load balancer.

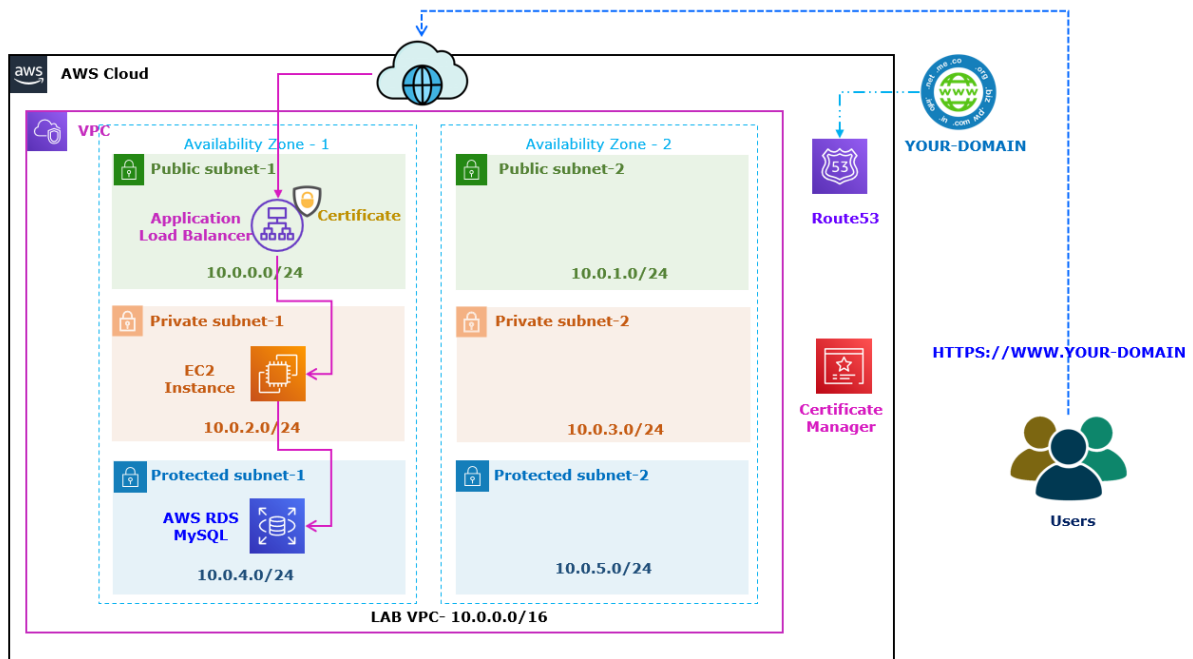
4) **Allow/ Deny**: Dropdown and select **Allow**.

Rule number	Type	Protocol	Port range	Source
100	HTTP (80)	TCP (6)	80	0.0.0.0/0
110	Custom TCP	TCP (6)	32768 - 65535	10.0.2.0/23
120	HTTPS (443)	TCP (6)	443	0.0.0.0/0
*	All traffic	All	All	0.0.0.0/0

5) Select **Save changes**.

Task 4: Access Secure Application

In this task you are accessing your application securely.



Step 1: Access the Load Balancer using HTTPS

28. From your **local Desktop/Laptop Web browser**, type **FQDN** of **www** **record** with **https** (like **https://www.ahmad.com**).

- Open the **Settings** (**gear icon** in the web page).
- You get the **Prompt** to enter the **Connection String**.
 - Type **Connection string** (which you have created in the previous step).
 - Select **Ok**.
- Add additional Product inventory** details.

Note: You can see the site is showing as **Secure (Lock sign)**.

The screenshot shows a web application titled "AWS Database LAB" with a server IP address of 10.0.2.146. On the left is a large purple sidebar. The main content area has a settings gear icon and three input fields: "Product Name" (containing "Speaker"), "Product Qty." (containing "80"), and "Product Price" (containing "890"). Below these is an "Add" button. At the bottom is a table with product details.

Product Name	Product Qty.	Product Price		
Mouse	33	330	✓	✗
Keyboard	50	450	✓	✗
LCD Monitor	40	7800	✓	✗
HDMI Cable	1150	300	✓	✗
Speaker	80	890	✓	✗

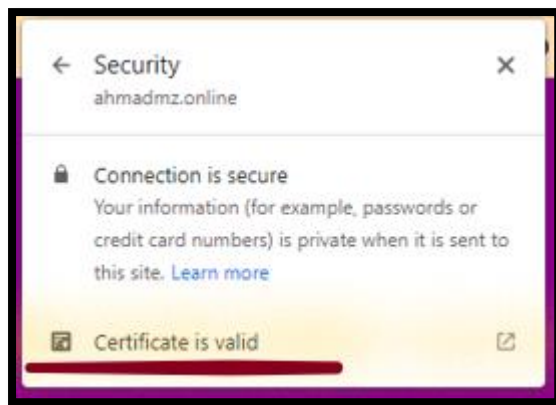
Step 5: View certificate Details

29. From **Webpage**, select **Lock sign**.

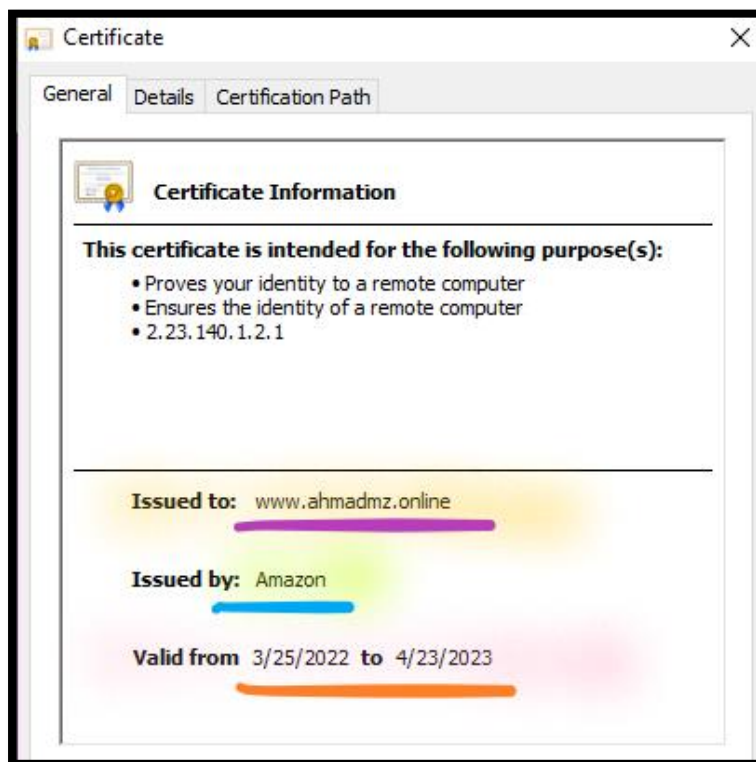
a. Select **Connection is secure**.



b. Select **Certificate is valid**.



Note: You can see the **Certificate** details.



c. Select **Ok**.

Step 6: Access the Load Balancer using HTTP

30. From your **local Desktop/Laptop Web browser**, type **FQDN** of **www record** with **http** (like **https://www.ahmad.com**).

Note: You can see the **site is accessible** with **http** and showing as **Not secure**.

Step 7: Forward HTTP Traffic to HTTPS

31. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

32. **Go to the left**, select **Load Balancers**.

33. Select **WebApp-LB**.

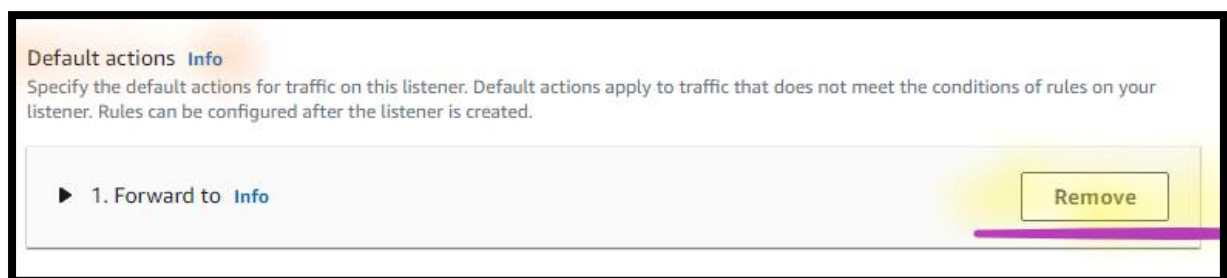
a. **Go below**, select the **Listeners**.

i. Select **HTTP :80 listener**.

ii. Select **Edit**.

a) In the **Listener details** section:

1) **Default actions**: Select **Remove**.



2) **Default actions**: Dropdown and select **Redirect**.

i. **Protocol**: Dropdown and select **HTTPS**.

ii. **Port**: Write **443**.

Default actions [Info](#)
Specify the default actions for traffic on this listener. Default actions apply to traffic that does not meet the conditions of rules on your listener. Rules can be configured after the listener is created.

▼ 1. Redirect [Info](#) Remove

Itemized URL **Full URL**

Protocol: **HTTPS** Port: **443**
1-65535 or to retain the original port enter #{port}

☒ Original host, path, query
Retain the host, path, and query from the request URL.

☐ Custom host, path, query
Modify any or all of these as needed for redirection.

Status code
301 - Permanently moved ▼

b) Select **Save changes**.

c) Select **View Listener**.

Note: You can see the **HTTP Rule redirecting** to **HTTPS**.

Listener ID	Security policy	SSL Certificate	Rules
☐ HTTP : 80 am...020e2d44a5b4da6 ▼	N/A	N/A	Default: redirecting to HTTPS://#{host}:443/#{path}?#{query} View/edit rules
☐ HTTPS : 443 am...b0d3b4cfef7f6125 ▼	ELBSecurityPolicy-2016-08	Default: c0a79dfb-9745-40ab-828b-1afe92d789a6 (ACM) View/edit certificates	Default: forwarding to LB-WebApp-TG View/edit rules

Step 8: Access the Load Balancer using HTTP

34. From your **local Desktop/Laptop Web browser**, type **FQDN** of **www record** with **http** (like **https://www.ahmad.com**).

Note: You can see the **site** is **Forwarded** to **https** and showing as **secure**.

Task 5: Cleanup the Environment

Step 1: Terminate an RDS

35. In the **AWS Management Console**, on the **Services** menu, click **RDS**.

36. Select the **Databases**.

- a. Select the **Inventory-DB** database.
 - i. Choose **Actions**.
 - ii. Select **Delete**.
 - iii. **Unselect** the **Create final snapshot**.
 - iv. Select the **I acknowledge**.
 - v. To confirm deletion, type **delete me**.
 - vi. Choose **Delete**.

Step 2: Terminate EC2 Instances

37. In the **AWS Management Console**, on the **Services** menu, click **EC2**.

38. Click **Instances**.

- a. Select the **WebApp Server** instances.
 - i. Click on **Instance state**.
 - ii. Select **Terminate instance**.
 - iii. Select **Terminate**.

Step 3: Terminate a Load Balancer

39. **From** the **EC2** Console:

40. Select the **Load Balancers**.

- a. Select **WebApp-LB** load balancer.
 - i. Choose **Actions**.
 - ii. Select **Delete**.
 - iii. Choose **Yes, delete** when prompted for confirmation.

41. Select the **Target group**.

a. Select **LB-WebApp-TG**.

i. Choose **Actions**.

ii. Select **Delete**.

iii. Choose **Yes, delete** when prompted for confirmation.

Step 4: Delete the Certificate

42. In the **AWS Management Console**, on the **Services** menu, click **Certificate Manager**.

43. Select **List certificate**.

a. Select **www Certificate**.

i. Select **Delete**.

ii. To confirm deletion, type **delete**.

iii. Choose **Delete**.

Step 5: Delete Security Groups

44. **From** the **VPC** Console:

45. Select the **Security groups**.

a. Select the **DB-SG**.

i. Select **Inbound rules**.

a) Select **Edit Inbound rules**.

1) **Delete all rules**.

2) Select **Save Rules**.

b. Select the **WebApp-SG**.

i. Select **Inbound rules**.

a) Select **Edit Inbound rules**.

1) **Delete all rules**.

- 2) Select **Save Rules**.
- ii. Select **Outbound rules**.
 - a) Select **Edit Outbound rules**.
 - 1) **Delete all rules**.
 - 2) Select **Save Rules**.
- c. Select the **LB-SG**.
 - i. Select **Inbound rules**.
 - a) Select **Edit Inbound rules**.
 - 1) **Delete all rules**.
 - 2) Select **Save Rules**.
 - ii. Select **Outbound rules**.
 - a) Select **Edit Outbound rules**.
 - 1) **Delete all rules**.
 - 2) Select **Save Rules**.
- d. Select the **LB-SG** and **WebApp-SG** and **DB-SG**.
 - i. Select **Actions**.
 - ii. Select **Delete security groups**.
 - iii. To confirm deletion, type **delete**.
 - iv. Choose **Delete**.

Step 6: Delete VPC

46. In the **AWS Management Console**, on the **Services** menu, click **CloudFormation**.
 - a. Select **Stack**.
 - b. Select **Sec-LABVPC**.
 - c. Select **Delete**.

- d. Select **Delete stack**.

Step 7: Release the EIP

47. In the **AWS Management Console**, on the **Services** menu, click **VPC**.

48. Select the **Elastic IPs**.

- a. Select **Elastic IP**.
 - i. Select **Actions**.
 - ii. Select **Release Elastic IP address**.
 - iii. Select **Release**.

Note: After release EIP screen should be blank.

Note

- Do not delete the Route 53 Hosted Zone.
- You will be using them in the next lab.